

Solutions – Quiz 2

ECON 320 – Introduction to Mathematical Economics

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Instructions

- Answer all questions. Show all steps clearly to receive full credit.
 - You may use diagrams where appropriate.
 - Total time: 20 minutes.
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1. Consider the system:

$$\begin{cases} x^2 + y + z = \alpha \\ xy - z = \beta \\ y^2 + z^2 = 1 \end{cases}$$

- (a) (2 pts) Write the Jacobian matrix of the system with respect to (x, y, z) .
- (b) (2 pts) State the condition under which the Implicit Function Theorem guarantees the existence of differentiable functions $(x(\alpha, \beta), y(\alpha, \beta), z(\alpha, \beta))$.
- (c) (3 pts) Derive an expression for $\frac{\partial(x, y, z)}{\partial\alpha}$ using Cramer's rule.

Solution: (a)

$$J(x, y, z) = \begin{bmatrix} 2x & 1 & 1 \\ y & x & -1 \\ 0 & 2y & 2z \end{bmatrix}.$$

(b) IFT condition: $\det J(x, y, z) \neq 0$ at the solution point. Compute:

$$\det J = 4x^2z + 4xy - 2yz + 2y^2.$$

(c) Differentiating wrt α :

$$J \begin{bmatrix} \partial x / \partial \alpha \\ \partial y / \partial \alpha \\ \partial z / \partial \alpha \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Cramer's rule:

$$\frac{\partial x}{\partial \alpha} = \frac{2(xz + y)}{\det J}, \quad \frac{\partial y}{\partial \alpha} = \frac{-2yz}{\det J}, \quad \frac{\partial z}{\partial \alpha} = \frac{2y^2}{\det J}.$$

2. Suppose the transformation is defined by:

$$u = x^2 + y^2 + z^2, \quad v = xy + yz + zx, \quad w = xyz.$$

(a) (2 pts) Construct the Jacobian $\frac{\partial(u, v, w)}{\partial(x, y, z)}$.

(b) (2 pts) Show that the determinant vanishes when at least two of (x, y, z) are equal.

(c) (2 pts) Explain the implication for whether the transformation is locally invertible.

Solution: (a)

$$K(x, y, z) = \begin{bmatrix} 2x & 2y & 2z \\ y + z & x + z & x + y \\ yz & xz & xy \end{bmatrix}.$$

(b) If $x = y$, columns 1 and 2 are identical, so $\det K = 0$. Similar for $y = z$ or $x = z$.

(c) Implication: the transformation $(x, y, z) \mapsto (u, v, w)$ is not locally invertible when any two variables are equal, since $\det K = 0$ there.

3. Let the system be:

$$\begin{cases} F_1(x, y, \theta) = x^2 + y^2 - \theta = 0 \\ F_2(x, y, \theta) = xy - 1 = 0 \end{cases}$$

(a) (2 pts) Verify that the Jacobian determinant with respect to (x, y) is nonzero at $(x, y) = (\sqrt{2}, 1/\sqrt{2}), \theta = 3$.

(b) (2 pts) Use the Implicit Function Theorem to argue for local differentiability of $(x(\theta), y(\theta))$.

(c) (3 pts) Compute $\frac{dx}{d\theta}, \frac{dy}{d\theta}$ at this point.

Solution: (a)

$$J(x, y) = \begin{bmatrix} 2x & 2y \\ y & x \end{bmatrix}, \quad \det J = 2(x^2 - y^2).$$

At $(\sqrt{2}, 1/\sqrt{2})$, $\det J = 3 \neq 0$.

(b) Since $\det J \neq 0$, IFT guarantees differentiable functions $x(\theta), y(\theta)$ near $\theta = 3$.

(c) Differentiating:

$$J \begin{bmatrix} dx/d\theta \\ dy/d\theta \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

By Cramer's rule:

$$\frac{dx}{d\theta} = \frac{x}{2(x^2 - y^2)}, \quad \frac{dy}{d\theta} = -\frac{y}{2(x^2 - y^2)}.$$

At $(\sqrt{2}, 1/\sqrt{2})$:

$$\frac{dx}{d\theta} = \frac{\sqrt{2}}{3}, \quad \frac{dy}{d\theta} = -\frac{1}{3\sqrt{2}}.$$